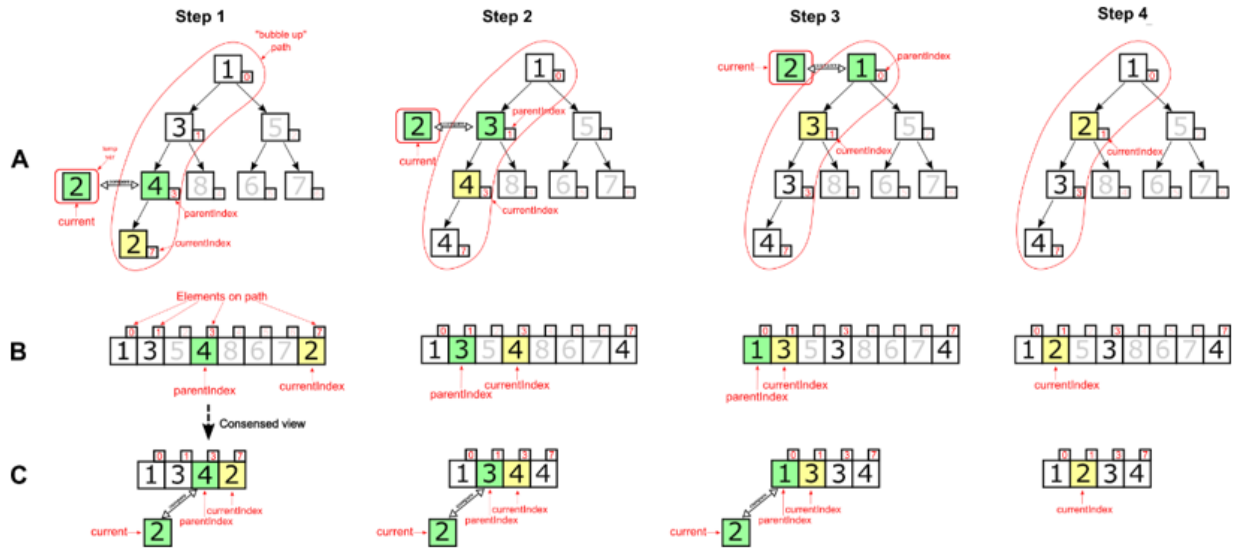


Figure 2.7. Bubbling up of an element in a min-heap, using the method that reduces the number of assignments to one third. The figure shows a call to `bubbleUp(pairs, 7)` displaying the tree representation (row A) and the array representation (B) for a min-heap (hence lower numbers bubble up to the top), and filtered elements (C) in the path `P` from the element to the root.

The first action needed is saving to a temporary variable the element `X` that will move up (step 1). Then, starting with its parent, we compare all elements in the path to the temporary variable, and copy them over in case they have lower priority (steps 1-3). At each time, we copy the parent element over the one child of its that is on the path `P`: it's like we filter out all the elements in the heap's array that are not part of `P` – this is highlighted in row C.

Finally, when we find an element `Y` along `P` that has higher priority than our temporary, we can copy